

Development of a Set of Best Practices for Distributed Pair Programming

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Abstract — This paper presents a research idea related to the use of Pair Programming in the context of Distributed Software Development. Organizations have been distributing their software development activities around the world for over a decade now which requires distributed teams to work together to achieve the project goal. Agile methods have been recently adopted by software engineers aiming to provide faster and more valuable results to the customer by fostering face-to-face communication, rapid response to changes, among others. Despite sounding contradicting, it has been claimed that distributed software development have been adopting agile methods as a strategy to make distributed teams more productive. Pair programming is one of the agile practices used in such context. It has been argued that pair programming in distributed software development contributes to increase the product quality and the productivity. For this reason, the goal of this idea proposal is to understand the advantages and challenges of Distributed Pair Programming, and to develop a set of best practices to facilitate its adoption.

Keywords— *Pair Programming, Distributed Software Development, Extreme Programming.*

I. DESCRIPTION OF THE PROBLEM AND JUSTIFICATION OF THE IDEA

Globalization has had an impact on different branches of industry, and consequently, in software development. In the scenario of seeking the most advantageous markets, lower cost of manpower, better product quality, among other factors, IT companies began to distribute their development software processes. In this context, in the late 90's Distributed Software Development (DSD) emerges. DSD is characterized as a software project that is developed by team members physically distant from each other [2].

Almost at the same time, with the publication of the Agile Manifesto in 2001, agile methods were made available with the main characteristic to be an adaptive approach toward environments with volatile requirements. For its innovative characteristic, agile methods have attracted interest from both academia and industry up to the present day [3], over a decade later. Extreme Programming (XP) [4] is considered one of the most-adopted agile method in industry [5]. It offers a range of techniques and practices with a focus on software development, unlike other methodologies such as Scrum [12] that has its focus more toward project management practices.

One of the most popular practices adopted by agile practitioners is called Pair Programming. Pair Programming (PP) is defined as the cooperation between two programmers using one computer to develop software. Although this concept has not been defined by the XP creators, it has gained popularity when XP was proposed. The benefits of adopting this practice are, among others: increased productivity, increased product quality due to the high number of defects found in the continuing review performed by the pair of developers, reduction of the risk of being dependent on only one person's knowledge about the code and module system, increased collaboration and team communication, and improvement of working conditions engaging programmers reducing their isolation feelings [1], [6]. Pair programming in a distributed environment, called Distributed Pair Programming, has become a common practice due to the availability of communication infrastructure.

With the trend of the DSD, some challenges regarding communication and collaboration emerge and Distributed Pair Programming (DPP) becomes an important source of research. Some studies have sought to investigate the effectiveness of the practice of DPP as shown in Baheti [7], Canfora [8] and Hanks [9]. Other studies, such as Natsu [10] and Stotts [11] proposed tools to support DPP. However, despite the investigations and discussions provided by these studies, there is no study to the best of our knowledge that has investigated how distributed pair programming is performed. Literature lacks studies that explore proposals for best practices to facilitate the performance of this activity in such distributed context. We believe that having knowledge about how DPP is conducted is relevant to the improvement of results achieved by this practice.

Therefore, the research question that guides this idea proposal is "What are the main benefits and challenges of the DPP and what best practices can enable a better adoption of the pair programming practice in distributed development?"

II. METHODOLOGY

In order to answer our research question, we will conduct an empirical study. We chose such approach because of the strong human intervention suffered by the execution of the practice. The research approach is going to be qualitative,

exploratory-type and is going to use primary and secondary research methods.

Broadly, we intend to adopt the following research methods: systematic mapping of the literature, field study, and experiment. The systematic mapping of the literature will guide the research of related work and research opportunities. The field study will collect information about how DPP is adopted in the industry. We will conduct interviews to identify needs and difficulties in day-to-day software development teams. We describe the goal of adopting each of these methods in more details below as a stepwise plan.

- To study existing literature aiming to improve knowledge related to agile and distributed software development.
- To systematically map related work aiming to identify a set of lessons learned already documented in literature. Findings from this step will be used to support the development of the subsequent research steps.
- To define a preliminary set of best practices for DPP based on the systematic mapping step.
- To conduct a field study field and experiment: consists in carrying out interviews with some software development companies that implement the DPP in order to evaluate the documented best practices and identify best practices that are not present in the preliminary set created.
- Experiment: The realization of an experiment is going to be planned to assess the benefits of a subset of these practices in a controlled environment, identifying possible areas for improvement.
- To define a final set of best practices for DPP based on the findings identified in the field study and in the experiment.

III. PLANNING AND MOTIVATION

This idea has been identified in the second semester of 2011 as a result of a investigation about the adoption of agile methods in distributed software development. This investigation was designed intended to deepen the theoretical knowledge in the area of agile methods in DSD aiming to identify research opportunities. The strategy was to identify articles of impact from the analysis of citations of these articles. The purpose of this idea was a result of opportunities arising in this analysis.

Another step towards the exploration of this theme was the experience as an intern in the development center of a large multinational organization located in the PUCRS Technology Park, called TECNOPUC, in Porto Alegre, Brazil. During the internship it was possible to observe how programmers dealing with distributed pair programming work. The findings from this experience motivated us to further investigate the topic.

We are currently concluding the systematic mapping aiming to identify related studies in the literature that describe lessons learned. As a consequence of this mapping, we will have a preliminary set of best practices for DPP. The design and planning of field study and of the experiment are also in course.

IV. POTENTIAL CONTRIBUTIONS

At the end of this study it is expected the results obtained could contribute to both the academia and industry. In academia, this research will contribute to furthering knowledge about how DPP takes place in practice and which best practices can help distributed teams to be more effective when they adopt this practice. This knowledge will add value to areas such as DSD, Agile Methods and Software Quality. Another contribution is the systematic mapping developed, which allow the knowledge of related studies and research opportunities for future work. The industry will benefit from this research from the set of best practices proposed of DPP, whose goal is to help organizations to identify the main needs of the difficulties in DPP, and have the support for the implementation of DPP more effectively and facilitated.

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